

Generative Artificial Intelligence

Sample educational document created for testing IntelliAssist AI.

1. What is Generative AI?

Generative Artificial Intelligence refers to AI systems capable of producing new content such as text, code, images, audio, or other structured outputs. Generative systems learn patterns from large datasets and use those patterns to generate outputs that are statistically consistent with the requested context. Large language models are a major example of generative AI for text and code.

2. Large Language Models

A Large Language Model (LLM) processes sequences of tokens and learns statistical relationships between them. During generation, the model predicts likely next tokens based on the preceding context. Modern LLM applications can support question answering, summarization, translation, coding assistance, and conversational interfaces. Their outputs should still be verified when accuracy is important.

3. Transformers and Attention

Most modern LLMs are based on transformer architectures. Self-attention enables the model to relate different parts of a sequence and build contextual representations. Transformer-based models can be scaled to large parameter counts and trained on substantial corpora, enabling broad language capabilities.

4. Prompt Engineering

Prompt engineering is the practice of designing instructions and context to guide a language model toward a desired output. Effective prompts can define the task, provide relevant context, specify constraints, request a particular format, and instruct the model about how to handle uncertainty. In RAG applications, retrieved document chunks are included as context in the prompt.

5. AI Assistants

An AI assistant combines a language model with application logic, tools, data sources, memory, and an interface. A document assistant extends a language model with document ingestion and retrieval so that users can ask questions about their own material. The model is responsible for language generation, while the surrounding application controls access to documents and retrieved evidence.

6. Risks and Limitations

Generative AI can produce incorrect or fabricated information, sometimes called hallucination. Models may also reflect biases in training data, expose sensitive information if systems are poorly designed, or generate insecure or inappropriate content. Responsible applications should constrain the model, protect user data, test outputs, and make uncertainty visible when appropriate.

7. Generative AI in IntelliAssist AI

In IntelliAssist AI, a Gemini or OpenAI API can serve as the generation layer. Retrieval supplies relevant document context, while the language model turns that context into a natural-language answer. Summarization can use the same generation capability. The system should instruct the model to rely on retrieved documents and to state when the available evidence does not contain an answer.